

MTH 446: Lecture # 8

Cliff Sun

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Lecture Span

1. Rigorous definition of continuity of complex functions

Missed the first two mins of Lecture, was just defining what a complex function is.

Functions of Complex variables

Example

Let $f(z) = z^2$ and $z \neq 0$, then $z = re^{i\theta}$, then

$$f(z) = r^2 e^{2i\theta} = r^2 (\cos 2\theta + i \sin 2\theta) \quad (1)$$

Then f doubles the angle parameter and the magnitude of the number.

Limits

Definition 0.1. (ϵ - δ definition of limit) Let $D \subseteq \mathbb{C}$ and z_0 be an accumulation point of D . Let $w_0 \in \mathbb{C}$ and $f : D \rightarrow \mathbb{C}$. We say that f has a limit w_0 as $z \rightarrow z_0$ as $\lim_{z \rightarrow z_0} f(z) = w_0$. In other words

$$\forall \epsilon > 0, \exists \delta_\epsilon = \delta_\epsilon(z_0, f) > 0 : z \in D, 0 < |z - z_0| < \delta_\epsilon \implies |f(z) - w_0| < \epsilon \quad (2)$$

Q: Why is it required that $0 < |z - z_0| \iff z \neq z_0$?

A: So we don't get singularities. It approaches, but it never equals. This is the fundamental definition of limits.

Q: Why does z_0 have to be an accumulation point?

A: Let $D = \mathbb{N}$ and $z_0 = 1$. We know that $z = 1$ is an isolated point of $\mathbb{N}$. Then, we don't find a point $0 < |z - 1| < \delta_\epsilon$ for some ϵ if $\delta_\epsilon < 1$. Otherwise, if z_0 isn't an accumulation point, then we choose $z = 1$. This allows us to define continuous limits.

Example

Let $f(z) = iz/2$. Prove that $\lim_{z \rightarrow 1} f(z)$ exists and find it.

Proof. We first have to find w_0 . Guess $w_0 = i/2$. Then, this means that for every $\epsilon > 0$, it follows that

$$\exists \delta > 0 : |z - 1| < \delta_\epsilon \implies |f(z) - i/2| < \epsilon \quad (3)$$

Then calculate

$$|i/2||z - 1| < \delta_\epsilon/2 = \epsilon \quad (4)$$

Then, here, $\delta_\epsilon = 2\epsilon$. Therefore, for every $\epsilon > 0$, choose $\delta = 2\epsilon$ such if $|z - 1| < \delta$, then it is the case that $|f(z) - i/2| < \epsilon$. $\square$

This means that if $\lim f$ exists, then no matter how z approaches z_0 , then $f(z)$ approaches w_0 . No matter the trajectory of which we approach z_0 , it must be the case that $f(z) \rightarrow w_0$.

Example

Let $f(z) = z/\bar{z}$. Prove that $\lim f$ does not exist.

Proof. Let $z = x \in \mathbb{R}$ without the origin. Then $z \rightarrow 0$. Then, consider $f(z) = x/x = 1$. Then, $f(z) \rightarrow 1$ as $z \rightarrow 0$. But on the complex plane, it is the case that $f(z) \rightarrow -1$ as $z \rightarrow i0$ if $z = iy$. Therefore, $\lim f$ DNE. $\square$

Algebraic properties of limits

Let $D \subseteq \mathbb{C}$ and z_0 be an accumulation point of D . Then, let $f, g : D \rightarrow \mathbb{C}$ and $c \in \mathbb{C}$. Then, suppose $\lim f$ and $\lim g$ exists, then

1. $\lim(cf) = c \lim f$
2. $\lim(f + g) = \lim f + \lim g$
3. $\lim fg = \lim f \lim g$
4. if $\lim g \neq 0$, then $\lim f/g = \lim f / \lim g$
5. In particular, $\lim 1/g = 1/\lim g$.

Example

If $\lim_{z \rightarrow z_0} z = z_0$. To solve this, we just use the definition of the limit for this problem. First,

$$|f(z) - z_0| < \epsilon \iff |z - z_0| < \epsilon \quad (5)$$

Choose $\delta = \epsilon$, then it is the case that if $|z - z_0| < \delta = \epsilon$, then $|f(z) - z_0| < \epsilon$.

Example

If $\lim_{z \rightarrow z_0} = \bar{z}_0$, then

$$|f(z) - \bar{z}_0| = |\bar{z} - \bar{z}_0| = |\overline{(z - z_0)}| = |z - z_0| < \epsilon \quad (6)$$

Therefore, if $\delta = \epsilon$, then $|z - z_0| < \delta = \epsilon$, then $|\bar{z} - \bar{z}_0| < \epsilon$.

Example

If $\lim_{z \rightarrow z_0} \operatorname{Re}(z) = \operatorname{Re}(z_0)$. Here, $\operatorname{Re}(z_0) = (z_0 + \bar{z}_0)/2$, then we use the properties of limits to obtain:

$$\lim \operatorname{Re}(z) = 1/2(\lim z + \lim \bar{z}) = 1/2(z_0 + \bar{z}_0) = \operatorname{Re}(z_0) \quad (7)$$

We can use this same technique for $\lim_{z \rightarrow z_0} \operatorname{Im}(z)$, and $\lim_{z \rightarrow z_0} |z|^2$ (use product of limits).

Example

$\lim_{z \rightarrow z_0} |z| = |z_0|$. Here, we use the definition of the limit.

$$|h(z) - |z_0|| < ||z| - |z_0|| \leq |z - z_0| < \epsilon \quad (8)$$

Then, choose $\delta = \epsilon$.

Continuity of functions of complex variables

Definition 0.2. Let $D \subseteq \mathbb{C}$ and z_0 is an accumulation point of D and $f : D \rightarrow \mathbb{C}$. Then, the function f is continuous if

1. $z_0 \in D = \text{dom}(f)$
2. $\lim_{z \rightarrow z_0} f(z)$ exists
3. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Hence, f is a continuous function at $z = z_0$ if f is defined at $z = z_0$ and $\forall \epsilon > 0, \exists \delta > 0$ such that $|z - z_0| < \delta$ and $z \in D$, then $|f(z) - f(z_0)| < \epsilon$.

f is called continuous in D if f is continuous for every point in D . The set of all functions f continuous in D is called $C(D)$.

We will review the algebraic properties of continuous functions next time.